First Progress Report

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GitHub : https://github.com/SandeshJIT/rl-maze-solver

1. What have you already achieved?

I was able to create a maze environment using DFS recursion with additional wall removal to create multiple paths and loops, making the maze more interesting for RL agents. The environment is built using Gymnasium and contains start, end, walls and traps. The goal of the agent is to reach the end given the starting point with as little steps as possible and avoiding any traps along the way. I have also implemented a Q-Learning based algorithm which can be used to train both single maze or multi maze setup and visualize its progress using evaluation metrics such as Cumulative Reward over training, Steps to reach exit, Maze Solve Rate and Exploration vs Exploitation. Additionally, I built an animated visualization that shows the trained agent solving the maze step by step. One key observation from Q-Learning is that it suffers from catastrophic forgetting as the agent trains on one maze and moves to the next, the previously learned Q-values get overwritten, and the agent must essentially retrain for each new maze.

2. What are your immediate next steps?

My immediate next step would be to play around with different reward systems and maze sizes to see how Q-Learning reacts to it. Later I want to start implementing Phase 2 of the project i.e. implementing Deep Q-Network (DQN) and Proximal Policy Optimization (PPO). DQN is a natural next step since it can take the full grid as input and distinguish between different maze layouts, potentially addressing Q-Learning's forgetting problem. PPO should handle sparse rewards well with its policy gradient approach.

3. Are there any challenges you are facing or adjustments you need to make to the project you initially proposed?

Till now I haven't faced any major hurdles or seen any major problem which would prevent me from completing the initial proposed goal. However, as suggested in the comments, I want to think about being able to add some additional challenges to the environment such as varying maze sizes, higher trap density, or dynamic obstacles to see how all 3 different algorithms react. However, I would first concentrate on finishing the core scope then explore additional things.

4. What is your overall plan to complete this project in the next month?

As mentioned, first I plan to refine the config for Q-Learning and then implement DQN and compare how much better/worse the algorithm would be at solving the mazes. Later I will introduce PPO and try to have a race kind of setup across a large number of mazes to analyze their performance. Finally, I want to build a visual which shows a race of all 3 agents trying to solve the same maze simultaneously.